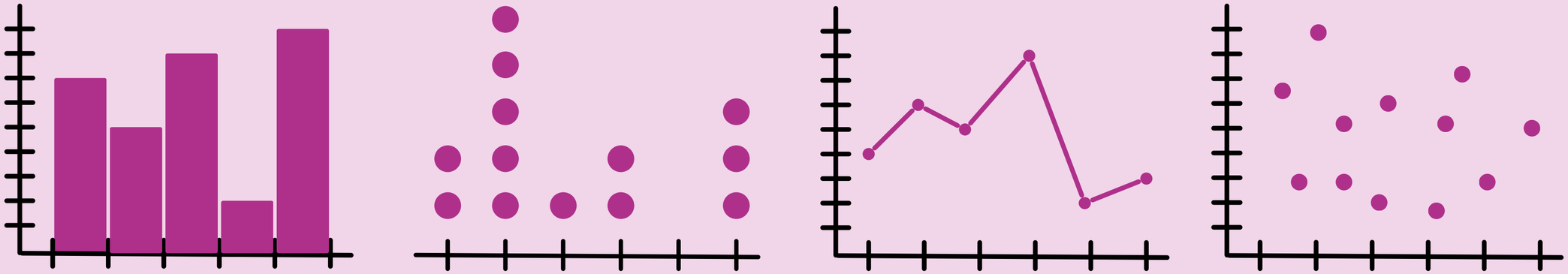
TELECOMMUNICATION

Customer Dataset(California)



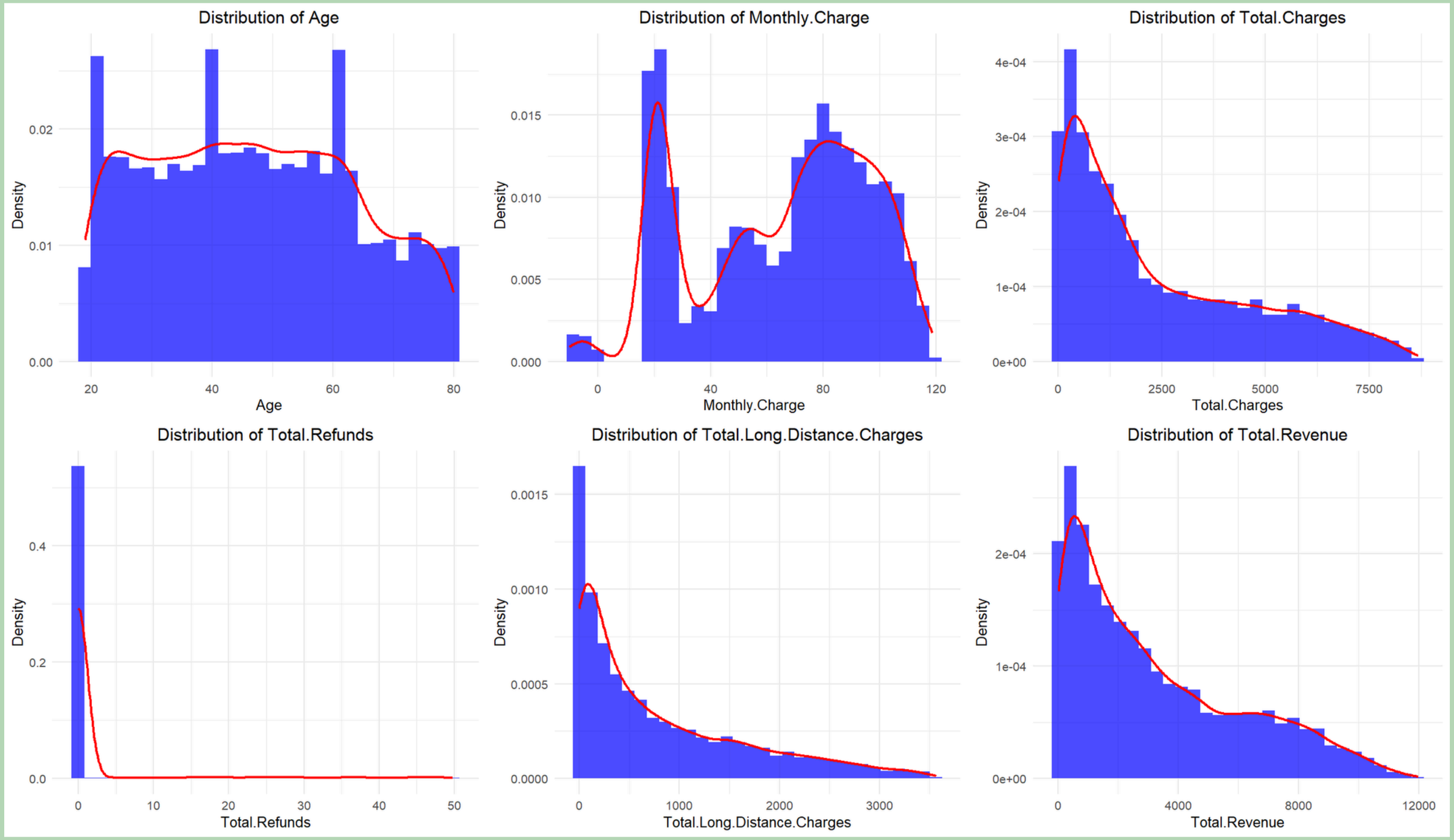
Cheng Liu & Chandana Yalavarthi
Professor: Paromita Guha

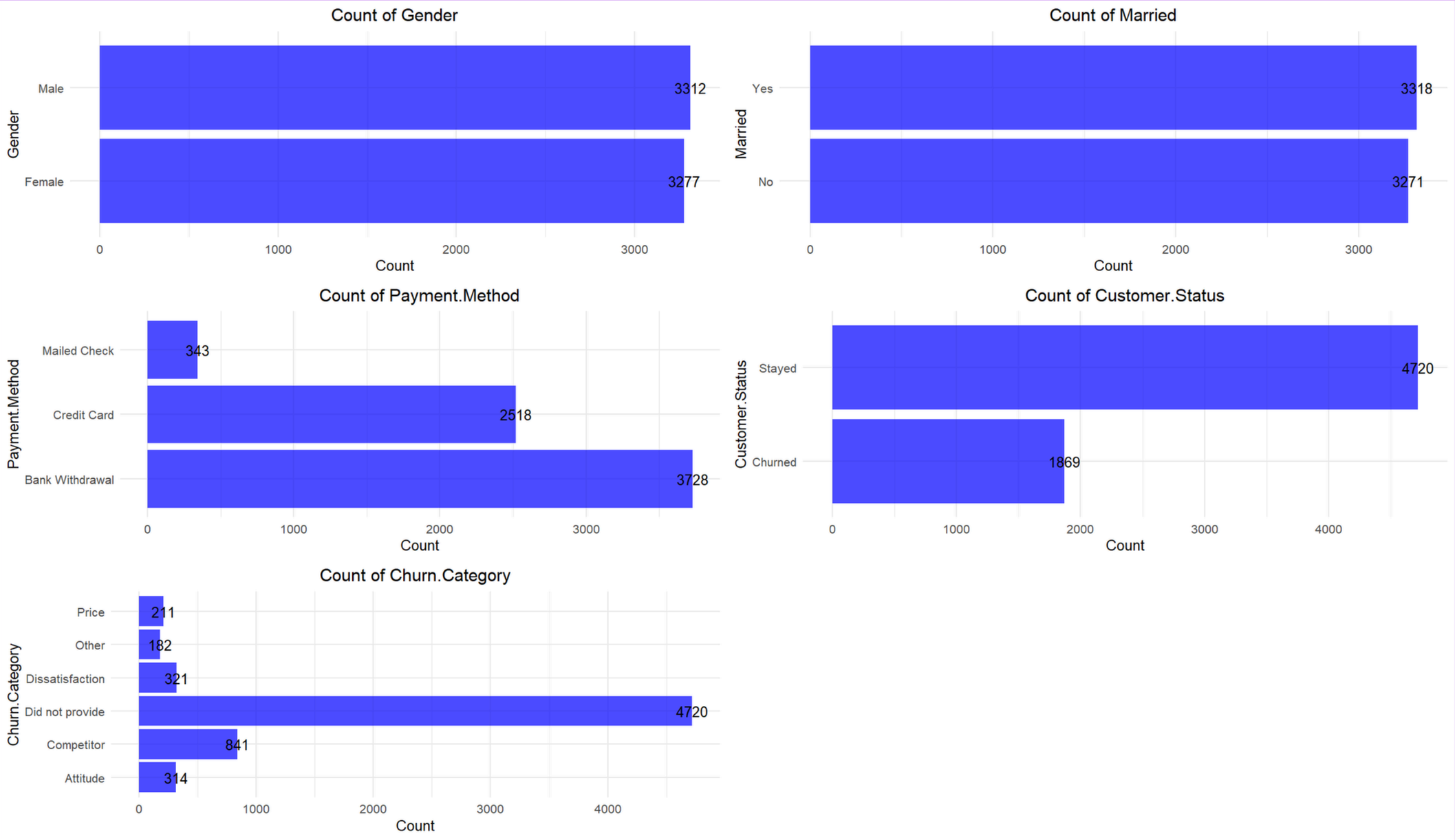
INTRODUCTION

- The data from the Customer Churn dataset was collected from a Telecommunications company in California in Q2 2022.
- The main goal of this project is to use the modeling methods we learned in this course to explore and predict customer churn.

Numerical Features		Categorical Features	
7,043 observations * 39 variables		Streaming Music	Streaming Movies
Age	Population	Customer ID	Unlimited Data
Number of Dependents	Total Refunds	Gender	Contract
Zip Code	Total Extra Data Charges	Married	Paperless Billing
Latitude	Total Long Distance Charges	City	Payment Method
Longitude	Total Revenue	Offer	Customer Status
Number of Referrals		Phone Service	Churn Category
Tenure in Months		Multiple Lines	Churn Reason
Avg Monthly Long Distance Charges		Internet Service	Device Protection Plan
Avg Monthly GB Download		Internet Type	Premium Tech Support
Monthly Charge		Online Security	Streaming TV
Total Charges		Online Backup	

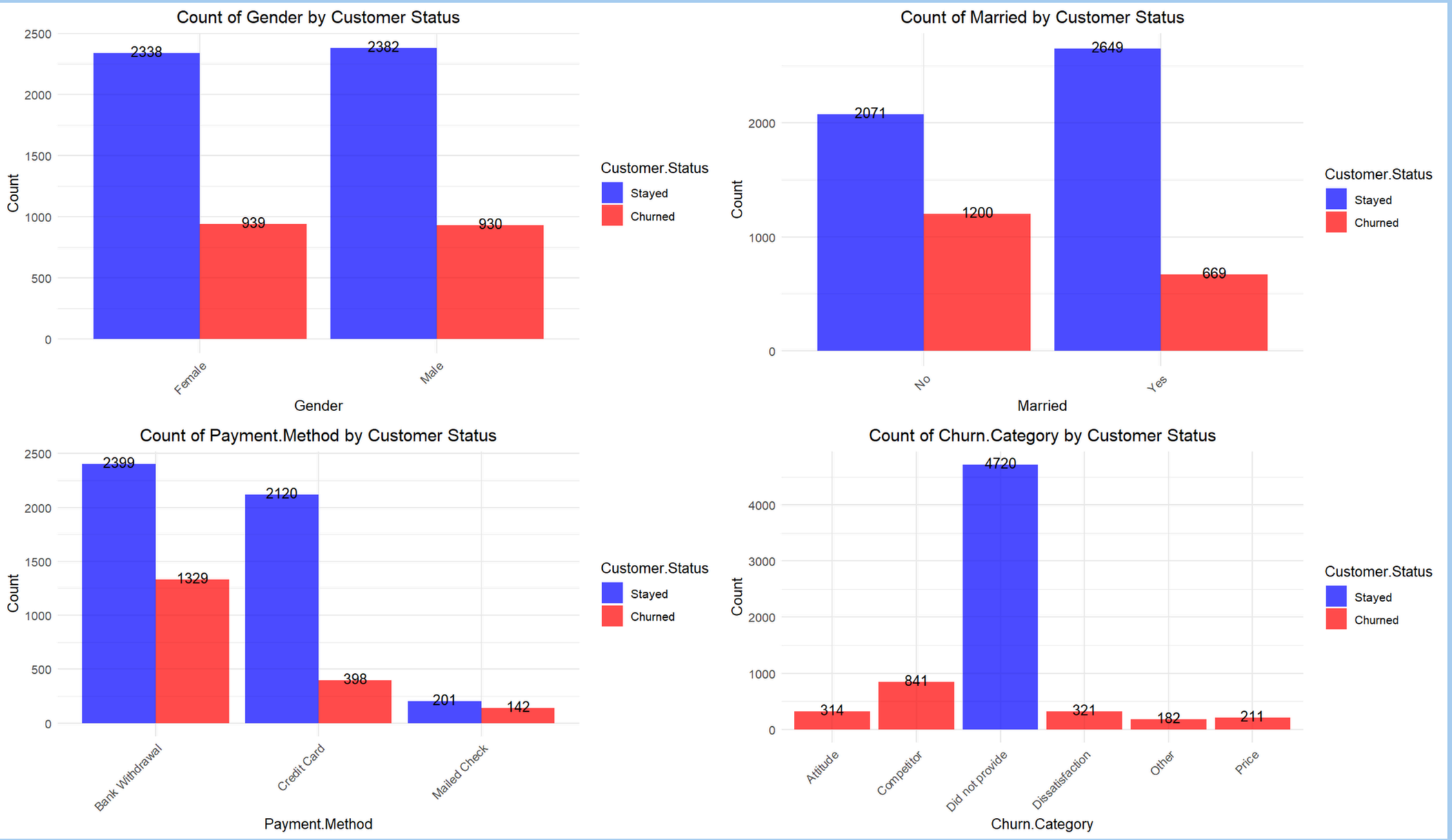
DISTRIBUTION OF NUMERICAL FEATURES

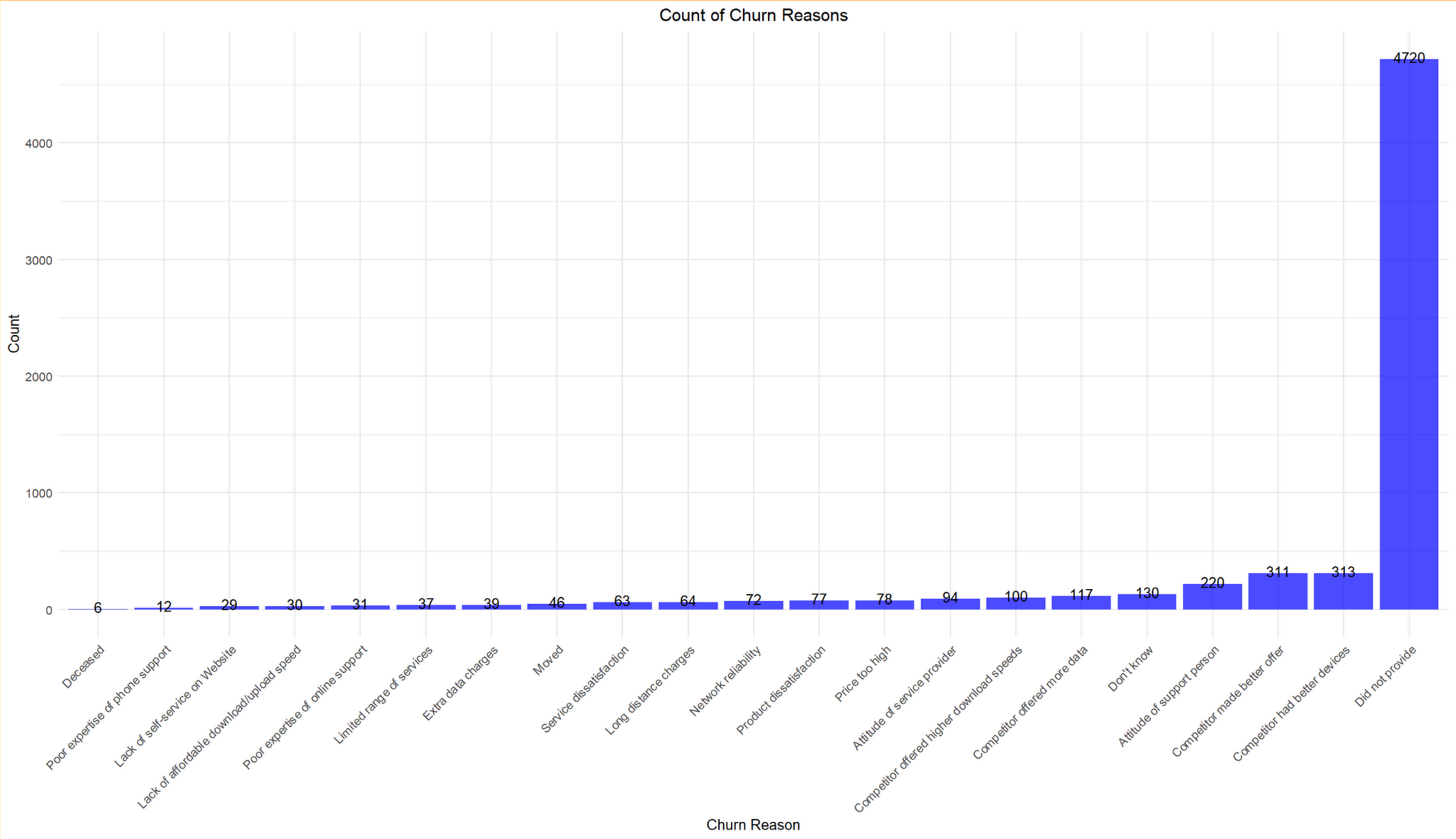




DISTRIBUTION OF CATEGORICAL FEATURES

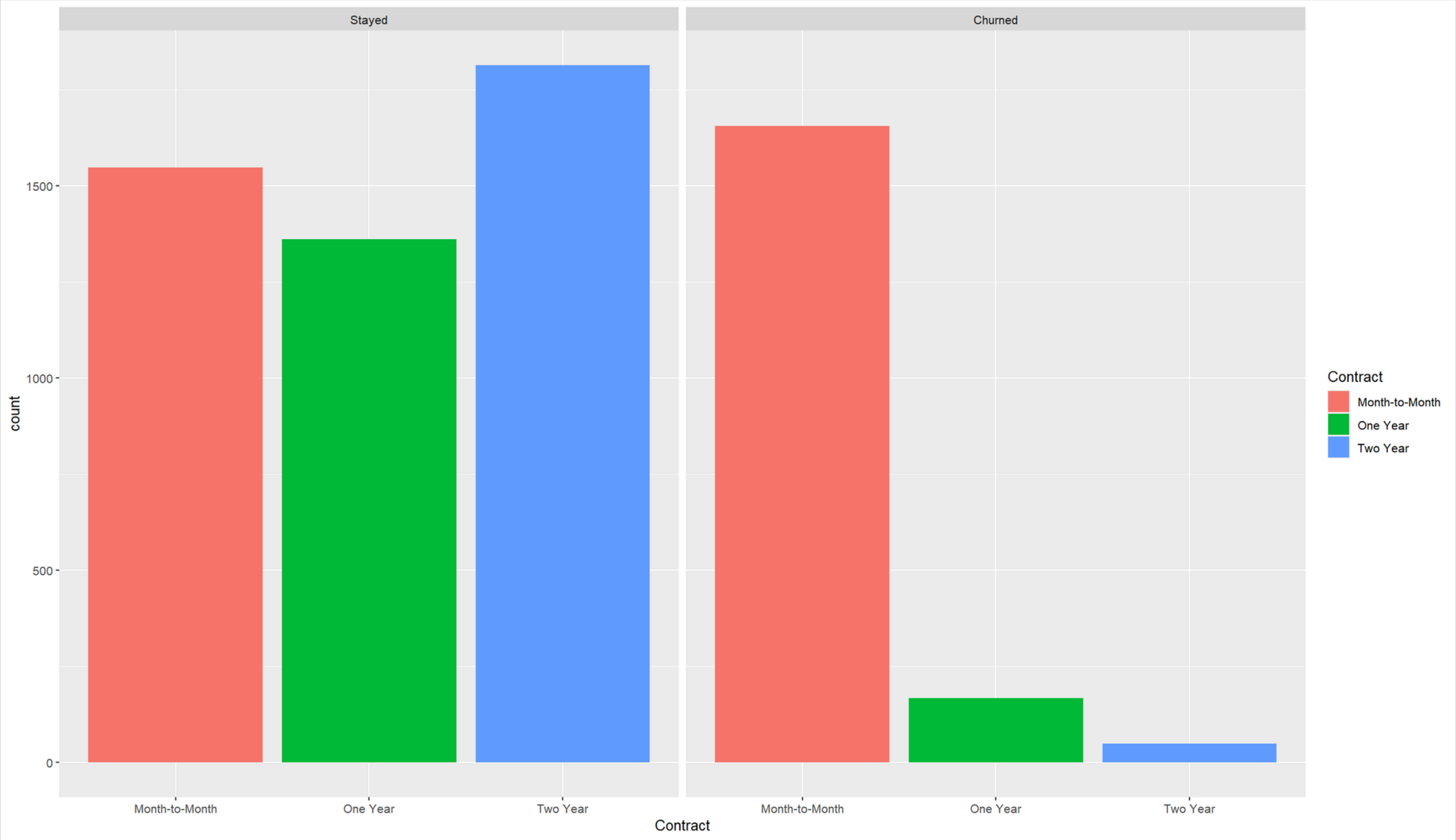
CATEGORICAL FEATURES BY CUSTOMER STATUS

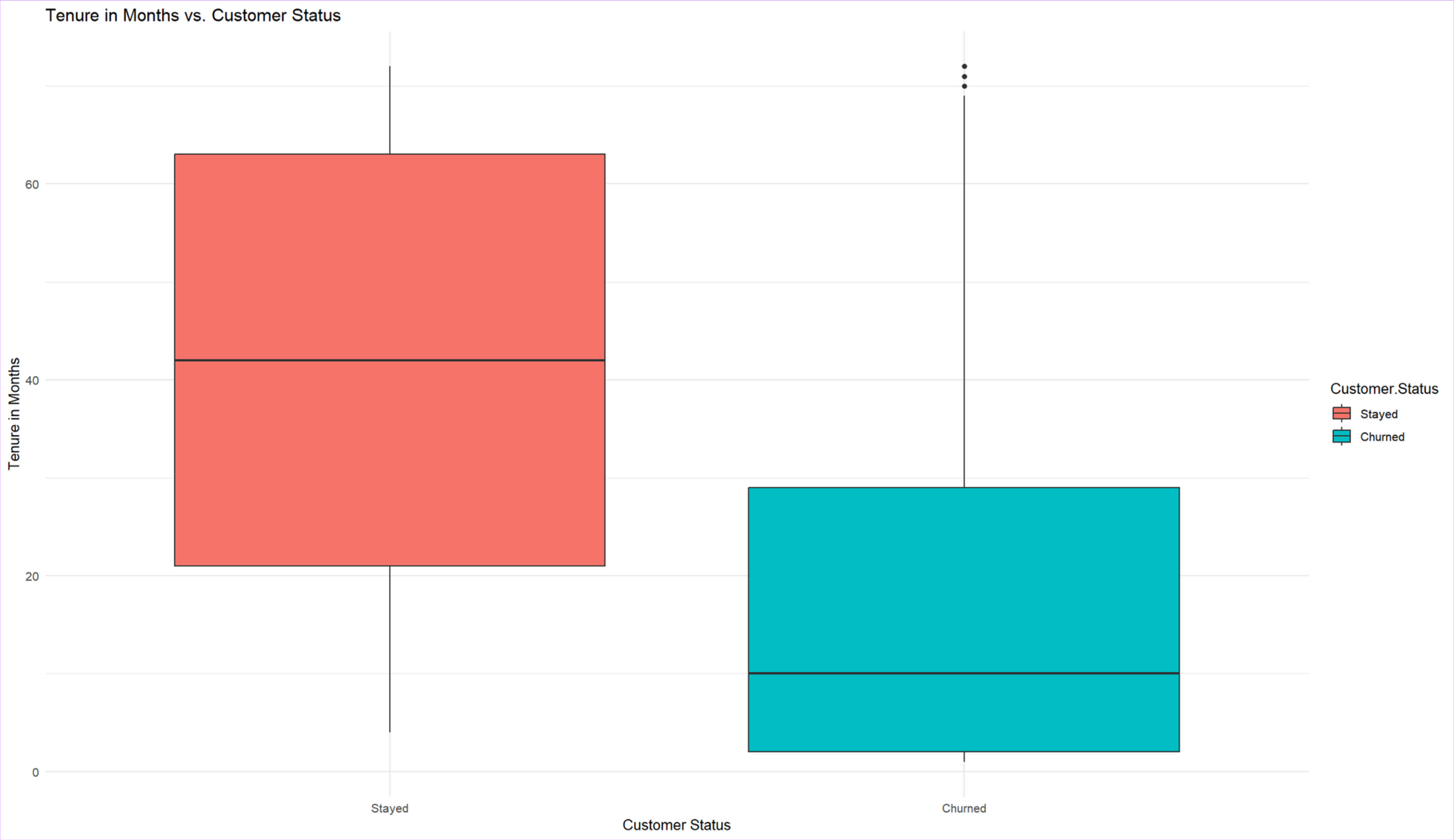




COUNT OF CHURN REASONS

CONTRACT VS. CUSTOMER STATUS



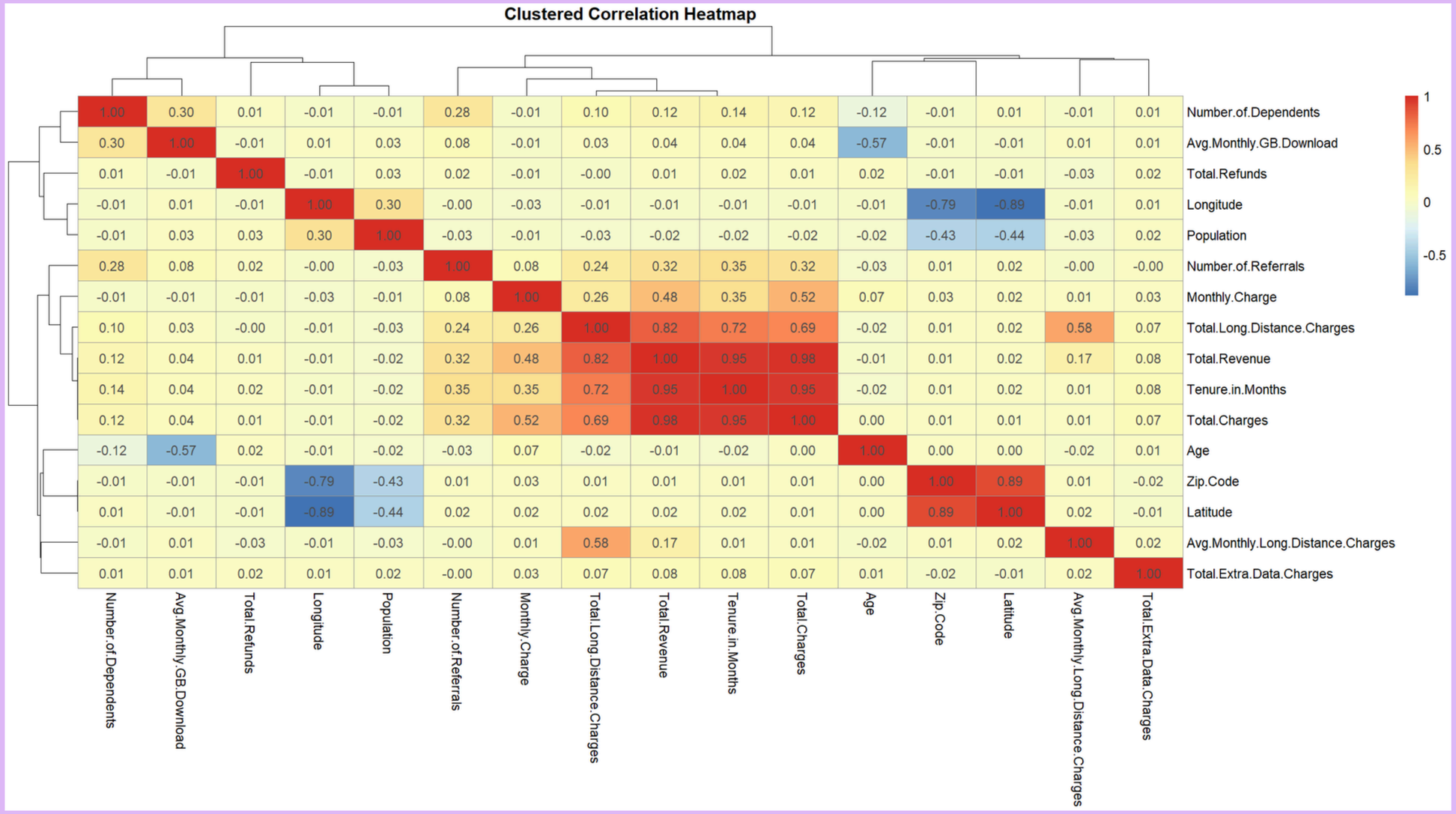


TENURE IN MONTHS VS. CUSTOMER STATUS

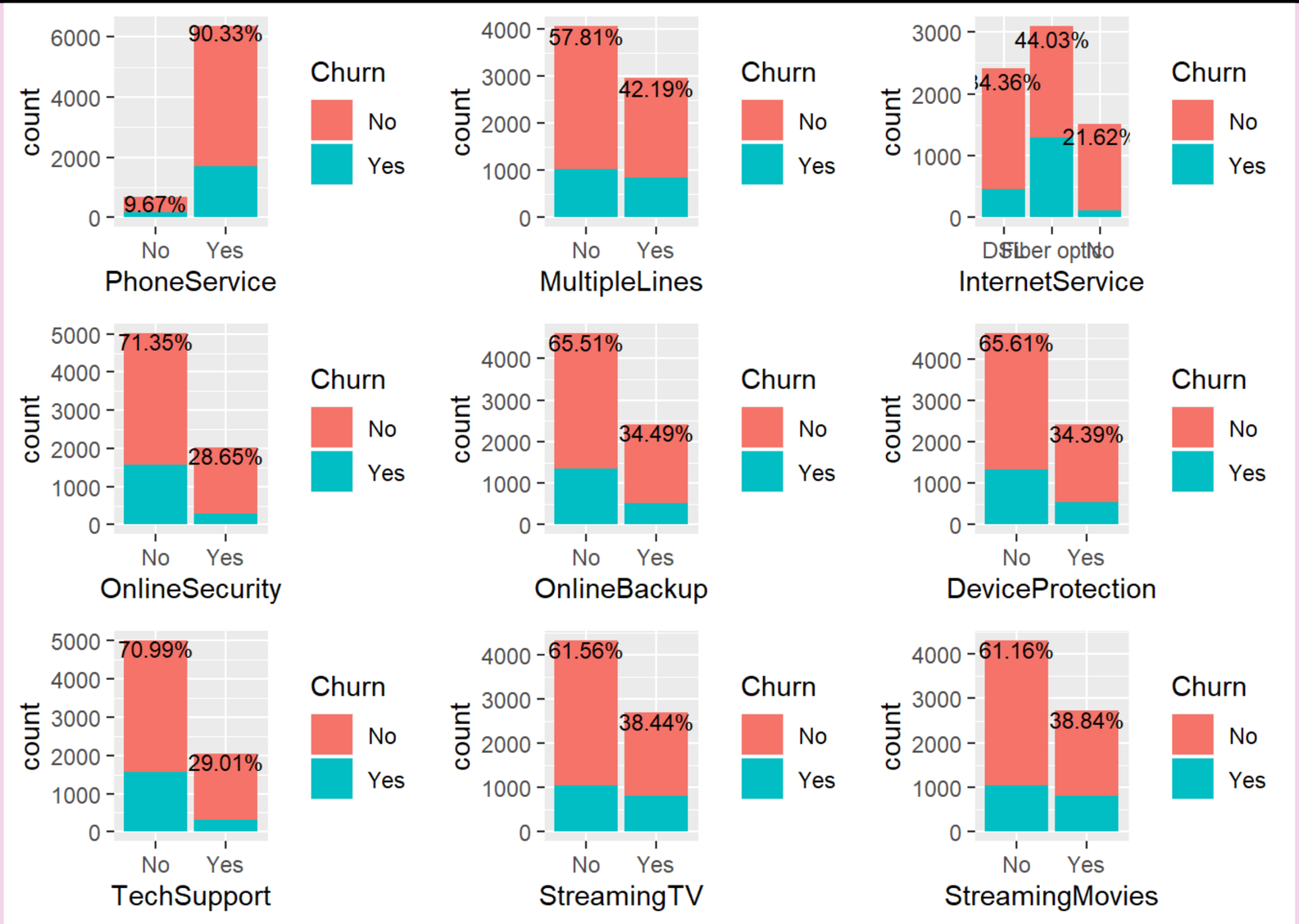
HEAT MAP

From this plot:

- **We can see that some variables like Total Charges/Revenue, and Tenure in Mon are very correlated.**
- **Total Long Distance Charges also show high correlation with Total Revenue/ Charges, cause higher usage typically comes with higher fees, thus increasing overall revenue.**



OFFERED SERVICES



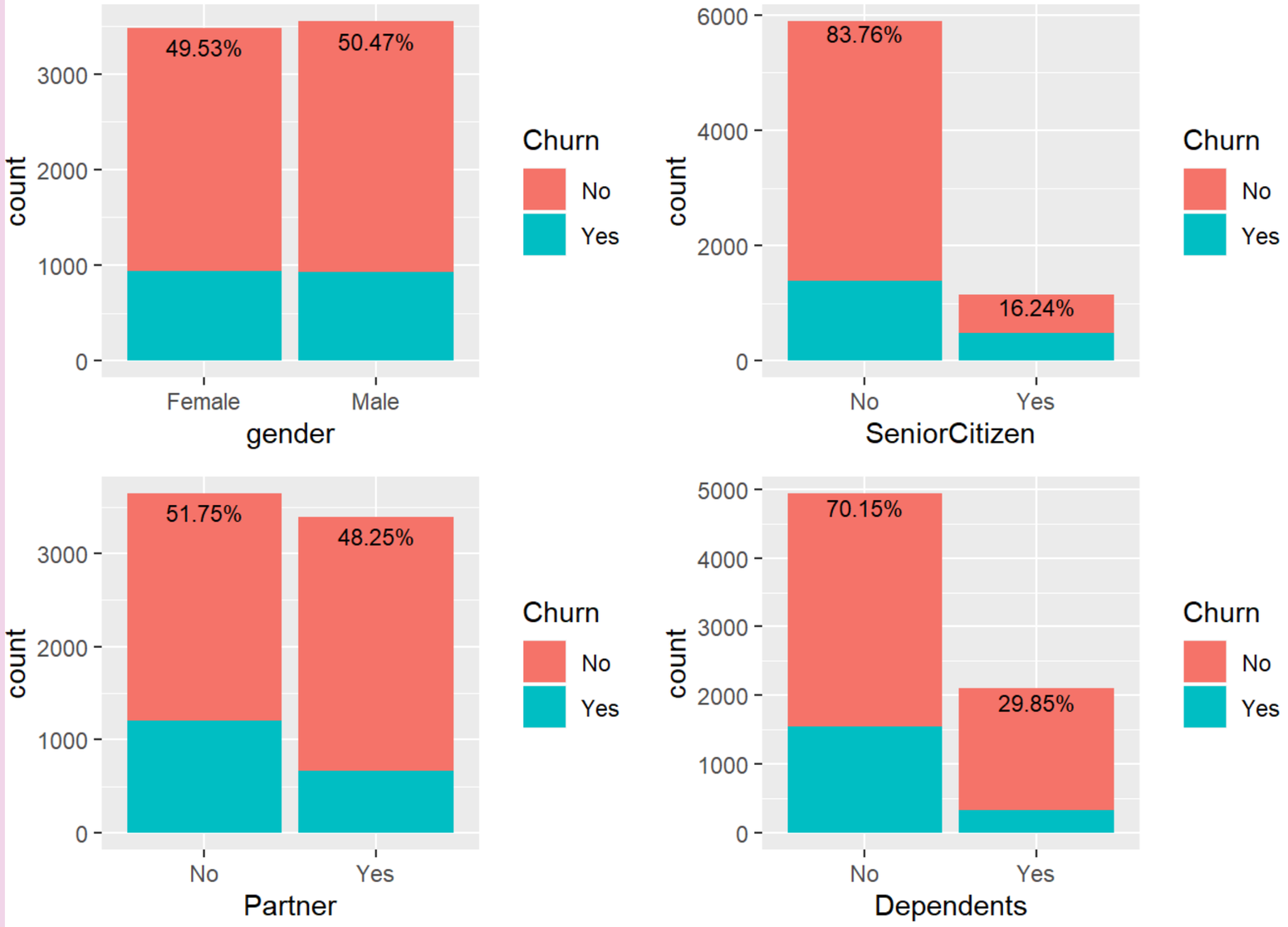
From the plot:

- **We can see that customers without additional services like online security, backup, device protection, tech support, and internet services are more likely to churn.**

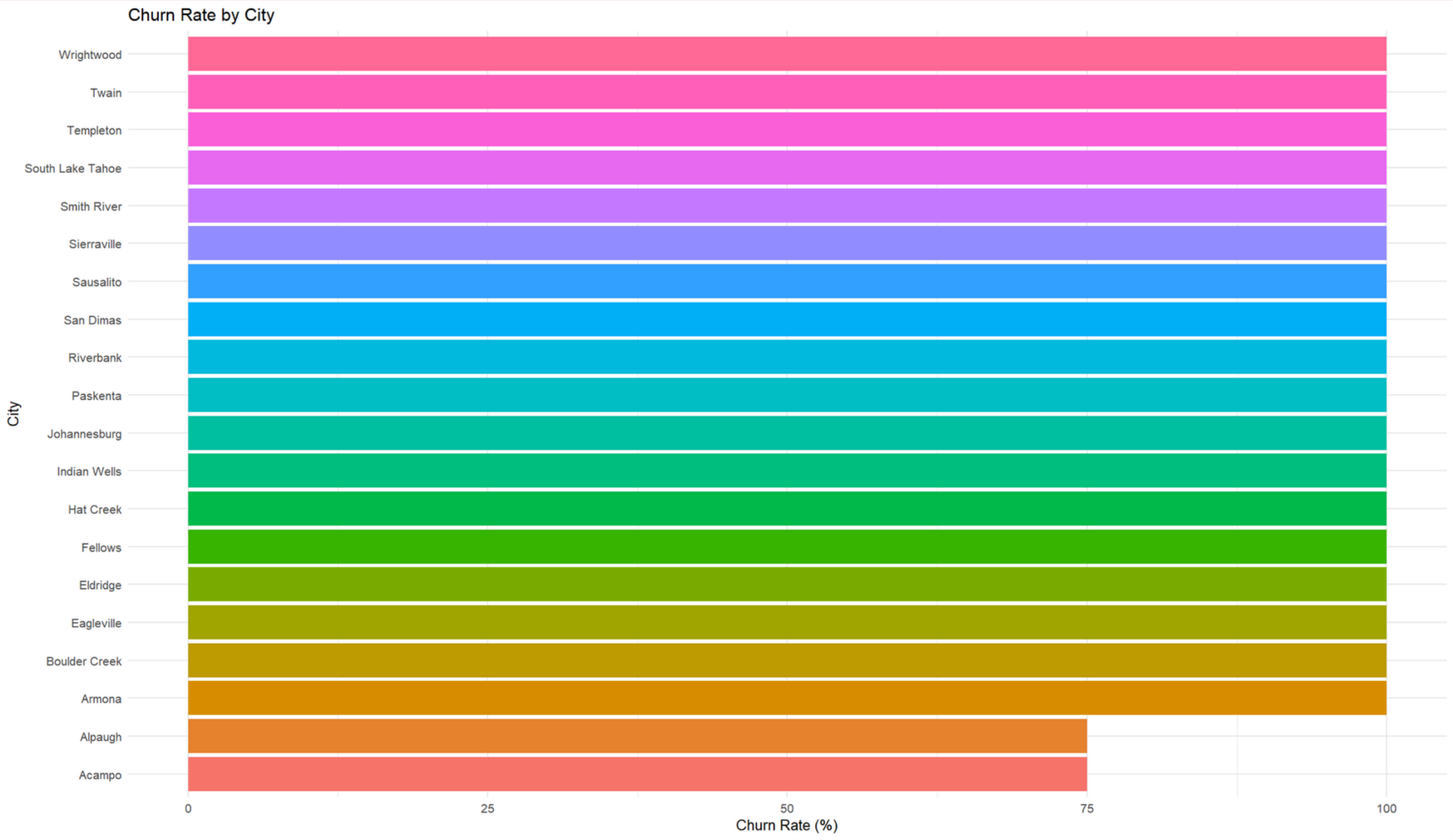
DEMOGRAPHIC DATA

From the demographic plot:

- **We can see that senior citizens, customers without dependents, partner are more likely to churn. However, gender show a more balanced distribution.**



CHURN RATE BY TOP 20 CITY



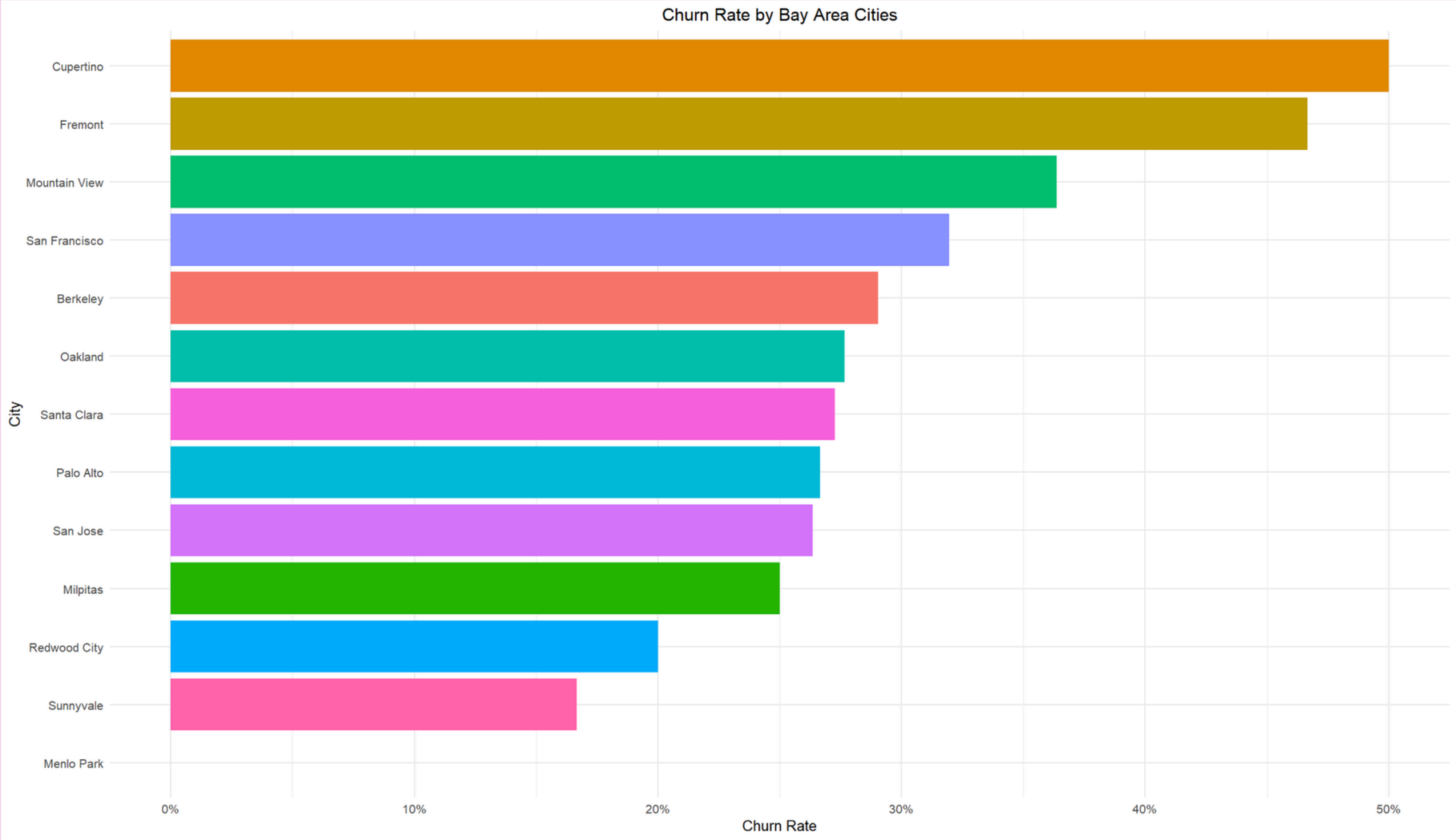
From the chart:

- **We can see that 18 cities all under 100% churn, which may mean there are no users in these area. Apart from the lack of data, telecommunications companies may be able to improve marketing or place advertisements in these places to develop new customer sources.**

CHURN RATE BY BAY AREA CITIES

From the chart:

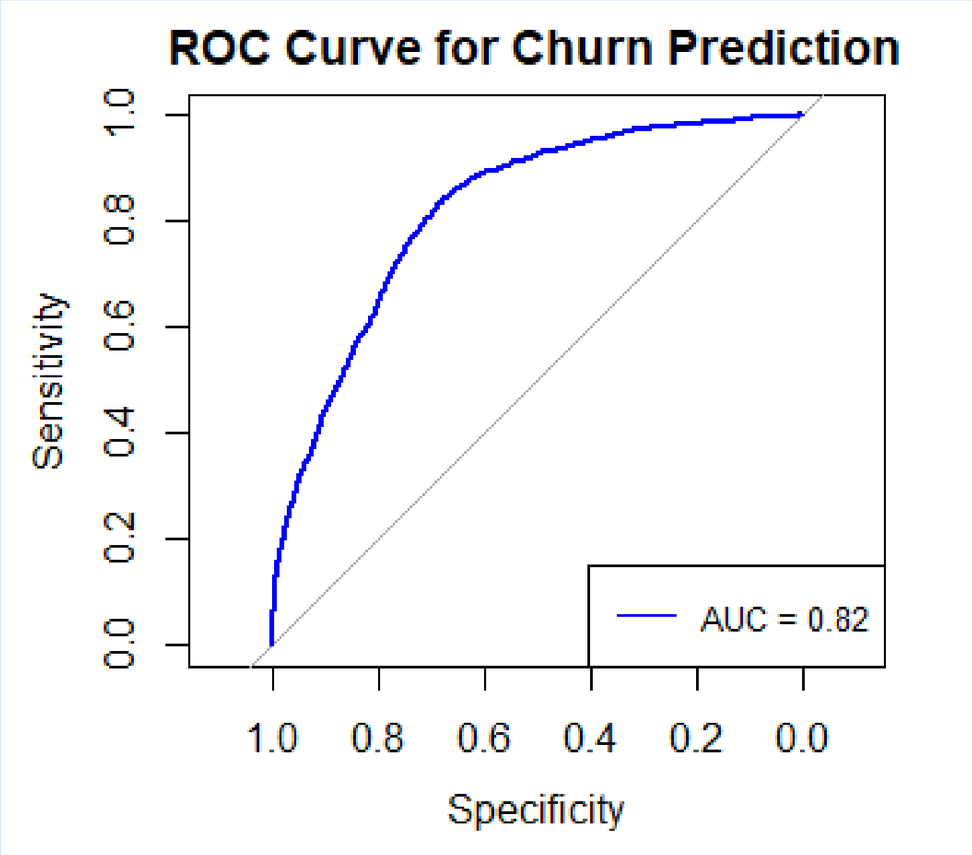
- **We can see the results for the top Bay Area cities. Due to insufficient or too small population data in some cities, the churn rate may be high. But it's still an indicator worth considering.**



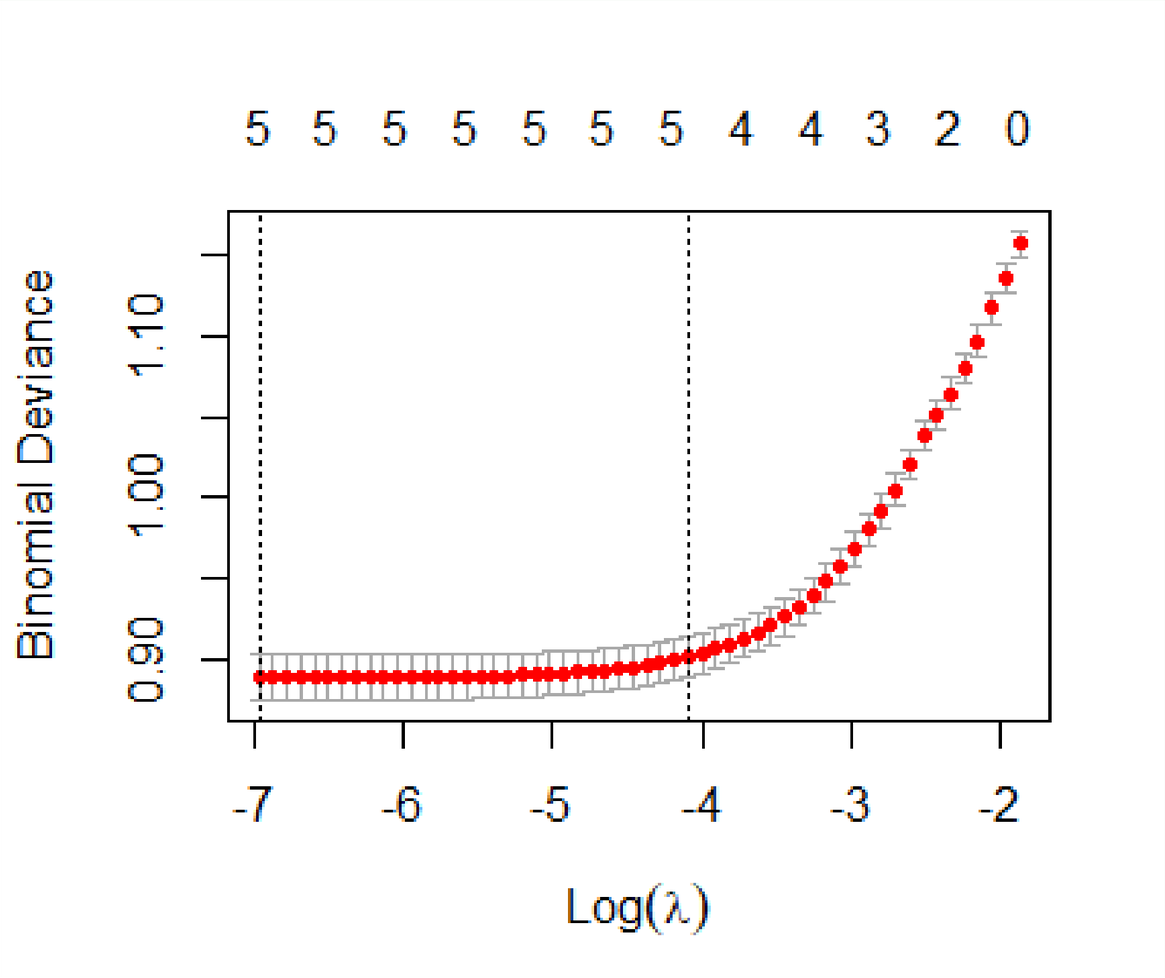
LOGISTIC REGRESSION

ROC (Receiver Operating Characteristic) & AUC (Area Under the Curve)

- X-axis (Specificity): It represents the True Negative Rate (TNR). It indicates the proportion of non-churning customers (negative class)
- Y-axis (Sensitivity): It represents the True Positive Rate (TPR). It indicates the proportion of churning customers (positive class)
- Here ROC curve hug the upper left corner of the graph. This indicates a model with 100% TPR and 100% TNR
- AUC of 0.82 means the model is good at distinguishing between customers who will churn and those who won't

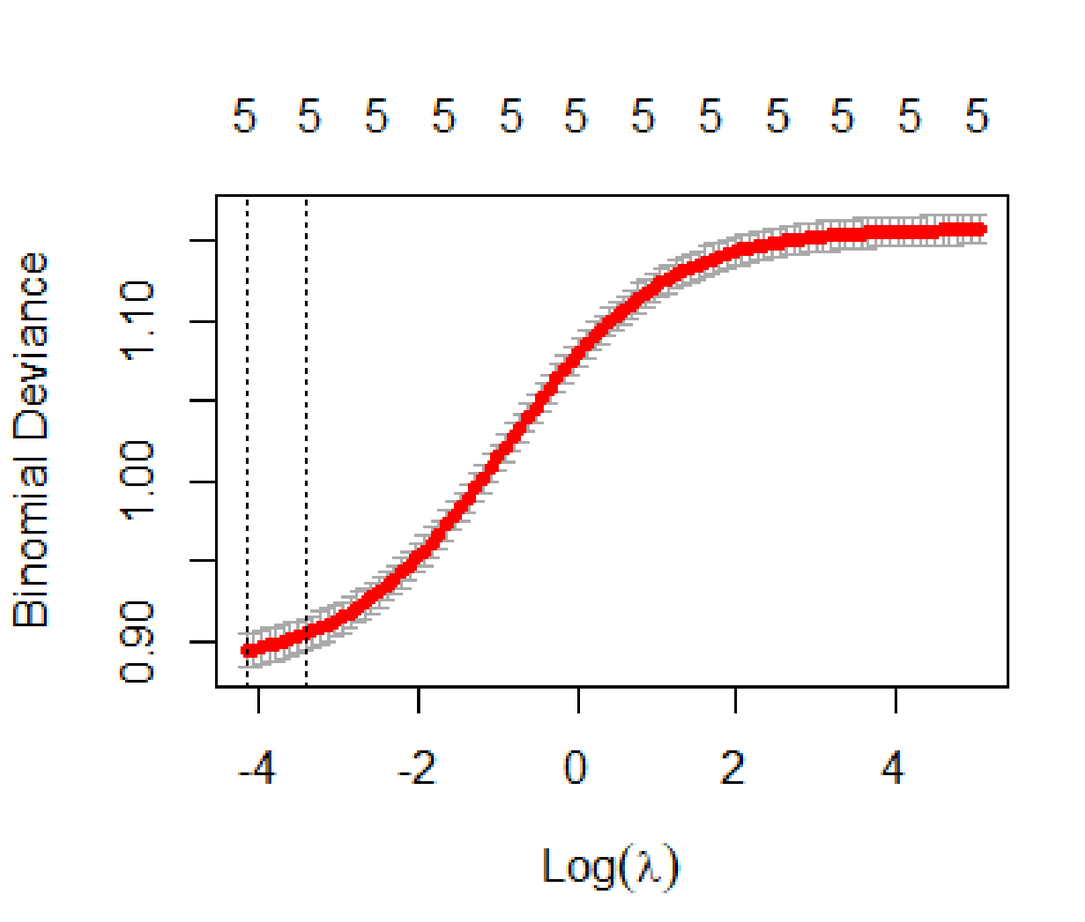


LASSO



- plot shows a characteristic "U-shape" in the deviance as λ
- λ increases when plotted on scale from -7 to -2.
- λ here is limited, focusing on smaller values, which tells us that the regularization strength needed isn't very high to achieve the optimal balance.

RIDGE



- Plot shows a gradual reduction in deviance as λ
- λ increases from -4 to around 2
- λ have little impact on the deviance,relatively stable across a broad range of λ

CONCLUSION

- Great for analyzing correlation between two variables-relationship between population density and churn rate .
- The logistic regression analysis using ROC (Receiver Operating Characteristic) and AUC (Area Under the Curve) indicates the model is good at distinguishing between customers who will churn and those who won't, with an AUC of 0.82.
- Overall, the dataset and analysis provide valuable insights into the relationships between customer demographics, geography, and churn rates, which can inform strategies to improve customer retention.

THANK YOU